

Implementing DOPPLER RADAR IN ADAS

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In previous issues of EFY, we explored the ADAS (advanced driver assistance system), which used lidar, camera, and ROS for

creation of the map. Now we are adding an important safety system for ADAS so that you can check the relative speed of moving vehicles.

Also, it detects the lane and marks the safe distance from the front vehicle in real time using the radar (see Fig. 1).

There are many lidar and radar systems available in the market for ADAS that can not only detect and map the environment but also tell us the speed of oncoming nearby vehicles as well as pedestrians, and accurately measure the distance for ADAS and autonomous driving systems. But since most of them are laser-based, they might not be very accurate in foggy or very low light conditions. So here we use the dop-

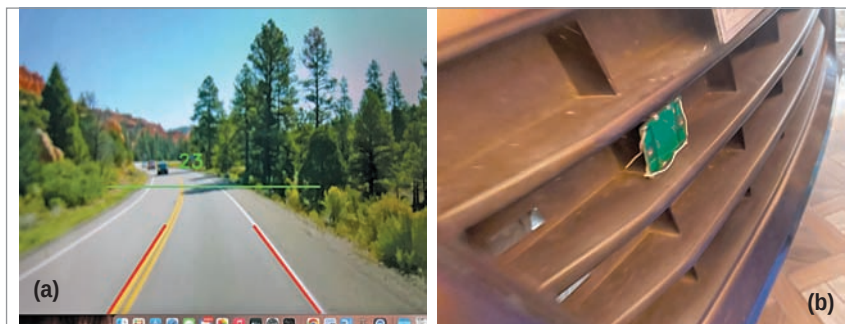


Fig. 1: (a) Radar and camera showing speed, direction, and lane, and (b) car with the doppler radar

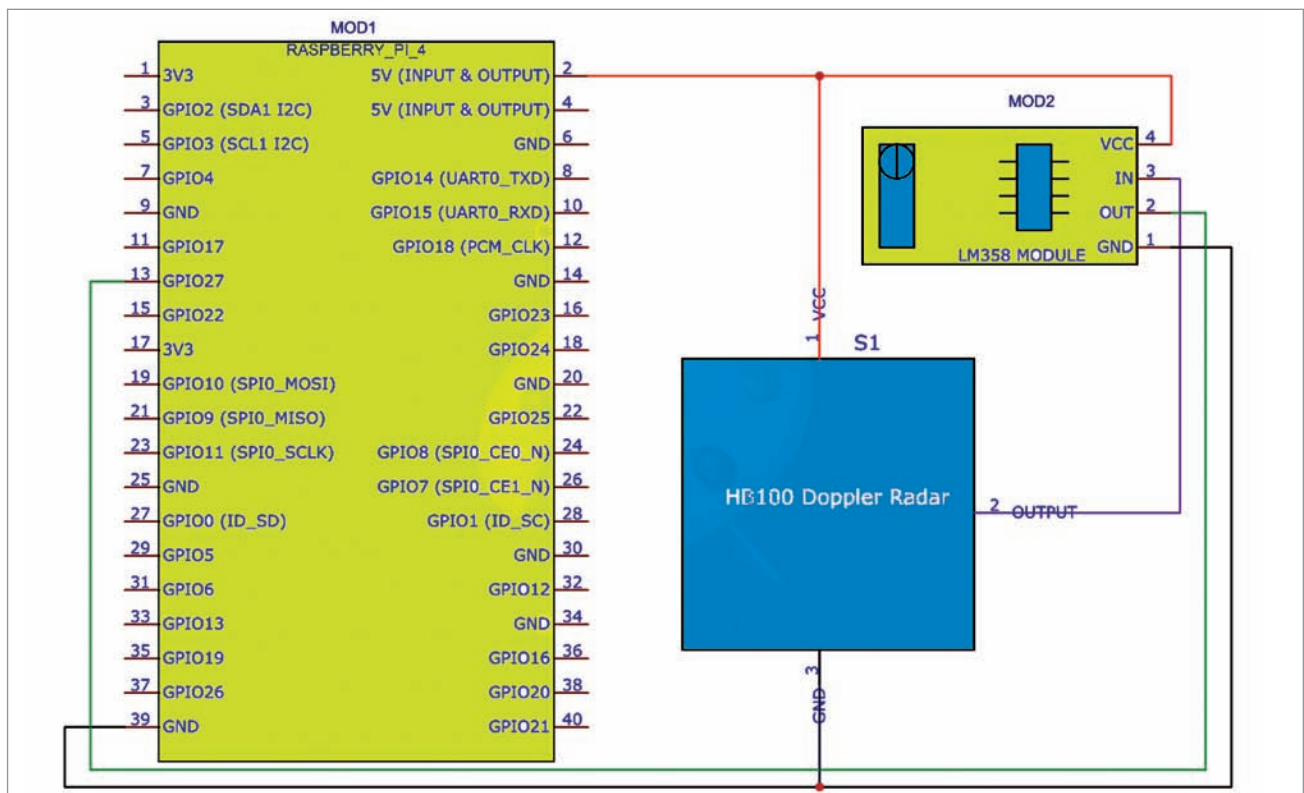


Fig. 2: Circuit diagram of the doppler radar in ADAS